2×2 Vedic Multiplier

VLSI Schematic Design · Cadence Virtuoso | gpdk090 90 nm CMOS

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Abstract

This project presents the complete transistor-level schematic design and functional simulation of a 2×2 Vedic multiplier using Cadence Virtuoso with the gpdk090 90 nm CMOS process. The design is based on the *Urdhva Tiryakbhyam* sutra of Vedic Mathematics, which allows all partial products to be generated simultaneously, minimising the critical path. Standard logic cells — OR, AND, XOR gates and a Full Adder — were each built from scratch at the transistor level, verified via dedicated testbenches, and then composed hierarchically into the final multiplier. Transient simulation in Cadence Spectre confirms correct 4-bit output across all 16 input combinations with a supply of 1.8 V.

1. Introduction

Multiplication is a fundamental arithmetic operation found in DSPs, microprocessors, AI accelerators, and communication ICs. Conventional ripple-carry array multipliers suffer from long carry-propagation delays. The Vedic *Urdhva Tiryakbhyam* algorithm generates all partial products in one step and accumulates them in a shallow adder tree, yielding a two-level critical path and a compact, regular structure that maps efficiently onto VLSI layouts.

2. Urdhva Tiryakbhyam Algorithm

For 2-bit inputs A = a_1a_0 and B = b_1b_0, the 4-bit product P = $p_3p_2p_1p_0$ is computed by the following parallel partial-product scheme:

Output	Boolean Expression	Hardware element
P[0]	a0 · b0	AND gate
P[1]	(a1·b0) XOR (a0·b1)	2× AND + XOR + carry C1
P[2]	(a1·b1) XOR C1	AND + XOR + carry C2
P[3]	C2	Carry wire

Table 1: Partial product decomposition — Urdhva Tiryakbhyam

3. Design Environment

Parameter	Value
EDA Tool	Cadence Virtuoso

Process (PDK)	gpdk090 — 90 nm CMOS
Supply Voltage	1.8 V
NMOS device	gpdk090_nmos1v W=120 nm, L=100 nm
PMOS device	gpdk090_pmos1v W=120 nm, L=100 nm
Simulator	Cadence Spectre (ADE L)
Input pulses	v1=0, v2=1.8 V, tr=10 ps
Logic style	Static complementary CMOS

Table 2: Design tool and technology parameters

4. Building Block Schematics

4.1 OR Gate

The CMOS OR gate uses a series PMOS pull-up pair (PM0–PM1) and parallel NMOS pull-down pair (NM0–NM1) to implement the NOR function, followed by an inverting output stage (PM2, NM2) to produce the true OR output. W/L = 120n/100n for all devices.

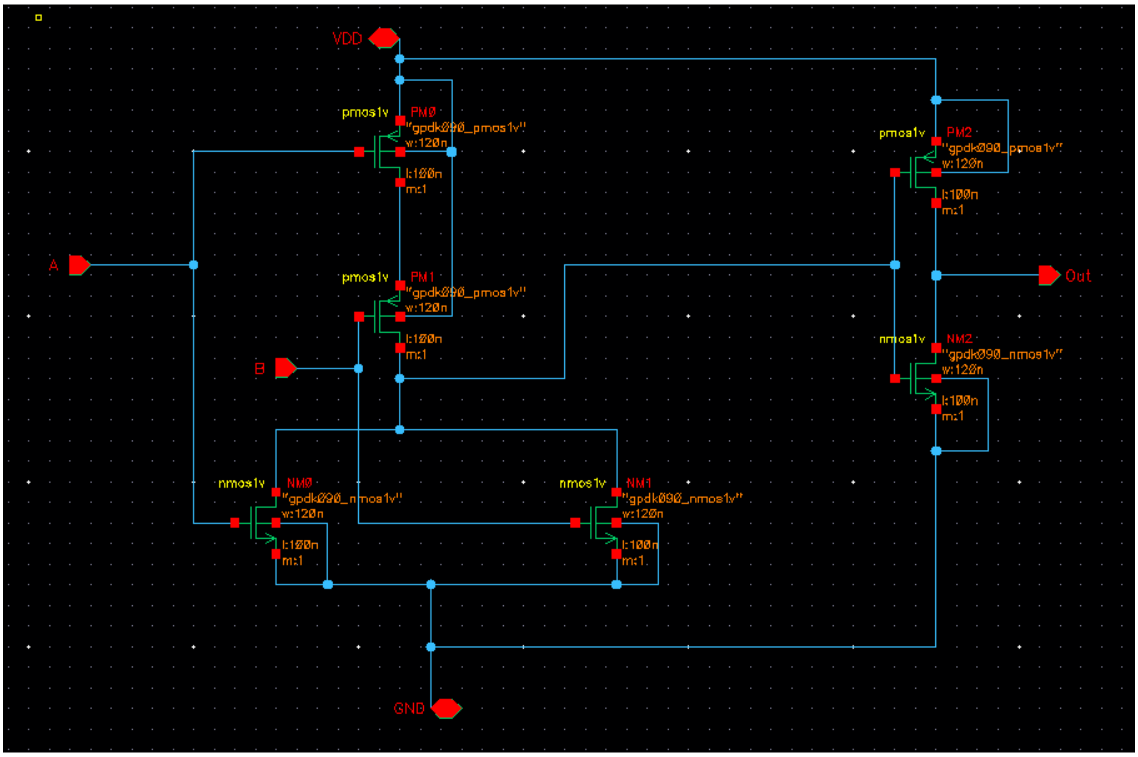

Figure 1: OR gate — transistor-level CMOS schematic (Cadence Virtuoso)

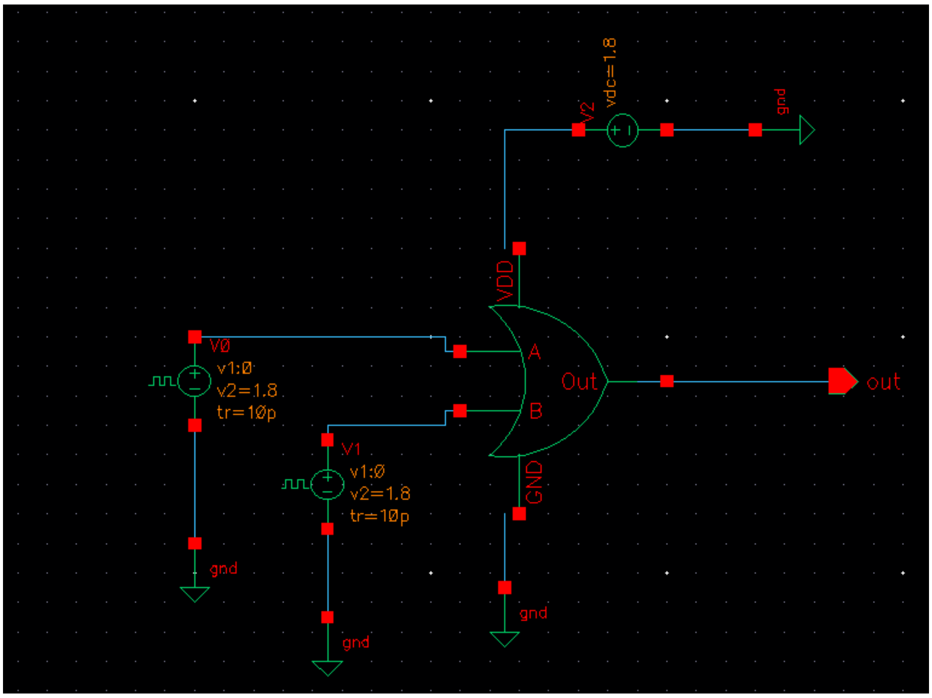

Figure 2: OR gate testbench — symbol with pulse sources V0, V1 and VDD = 1.8 V

4.2 AND Gate

The AND gate is implemented as a CMOS NAND (parallel PMOS pull-up PM0–PM1; series NMOS pull-down NM0–NM1) followed by a CMOS inverter output stage (PM2, NM2), producing Out = A AND B.

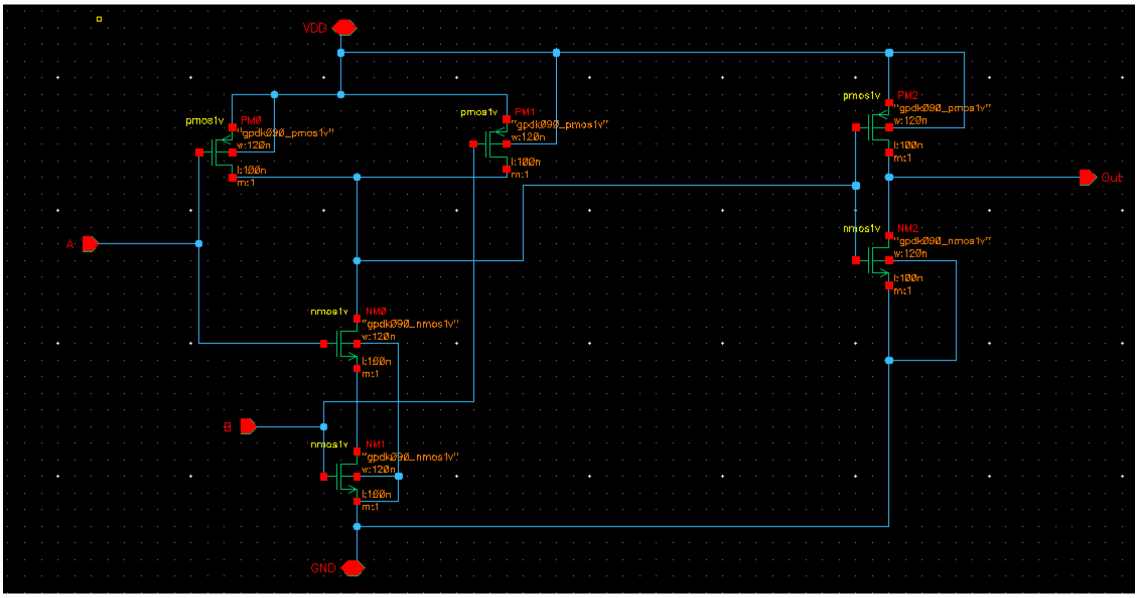

Figure 3: AND gate — transistor-level CMOS schematic

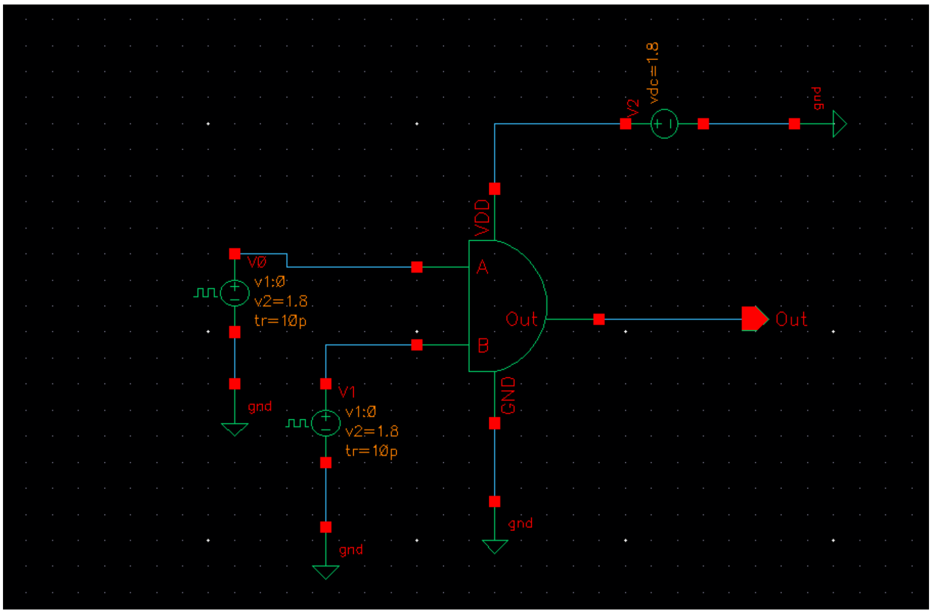

Figure 4: AND gate testbench

4.3 XOR Gate

The XOR gate uses a transmission-gate / complementary pass-logic topology. Inverted signals AP and BP are generated internally; the XOR output is formed by combining direct and complemented paths so that Out = A XOR B. The schematic also shows the AP/BP buffering inverters used to drive the pass network.

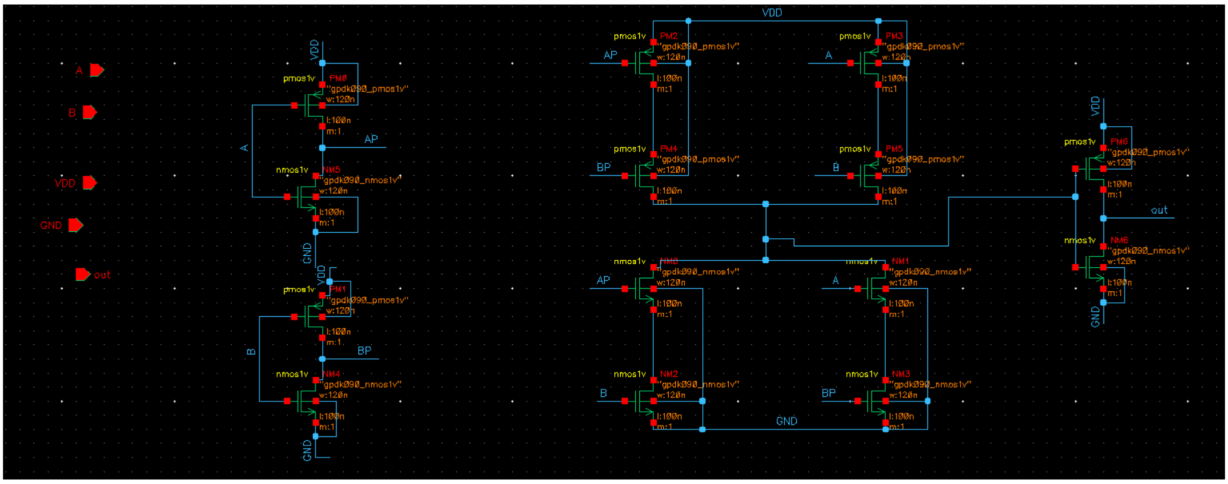

Figure 5: XOR gate — transistor-level schematic with complementary inputs AP, BP

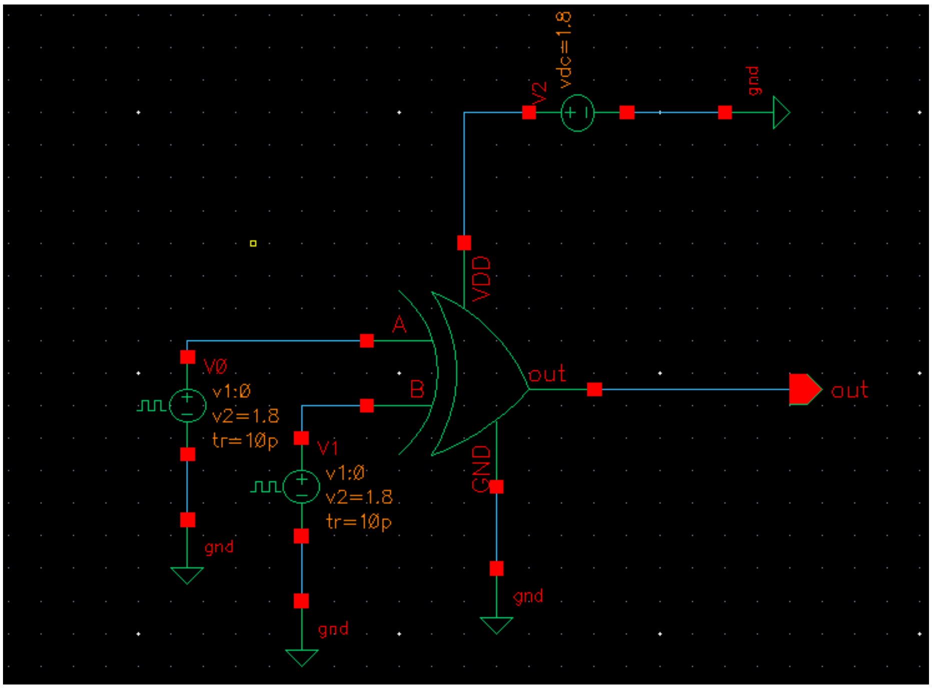

Figure 6: XOR gate symbol-level testbench

4.4 Full Adder

The Full Adder is composed hierarchically from the XOR and AND gate symbols. It accepts inputs A, B, and Cin and produces SUM = A XOR B XOR Cin and CARRY = AB + BCin + ACin. Two XOR gates handle the sum path; AND and OR gates handle carry generation.

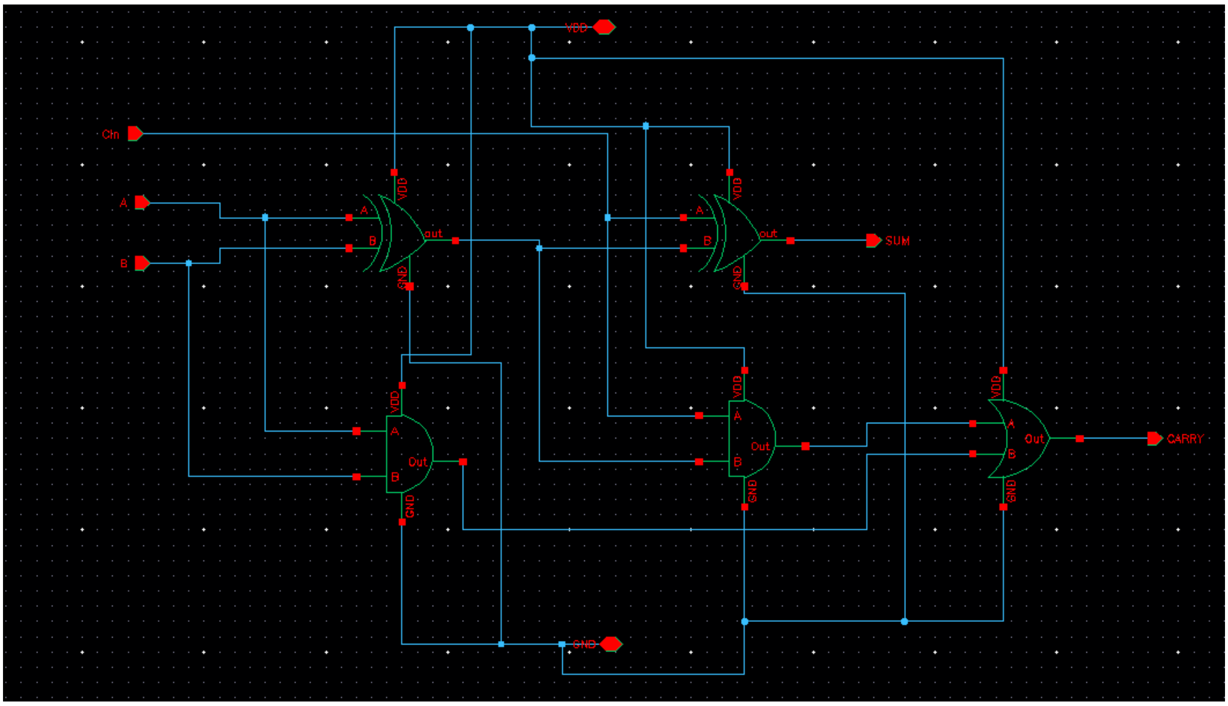

Figure 7: Full Adder — hierarchical schematic using XOR / AND gate symbols

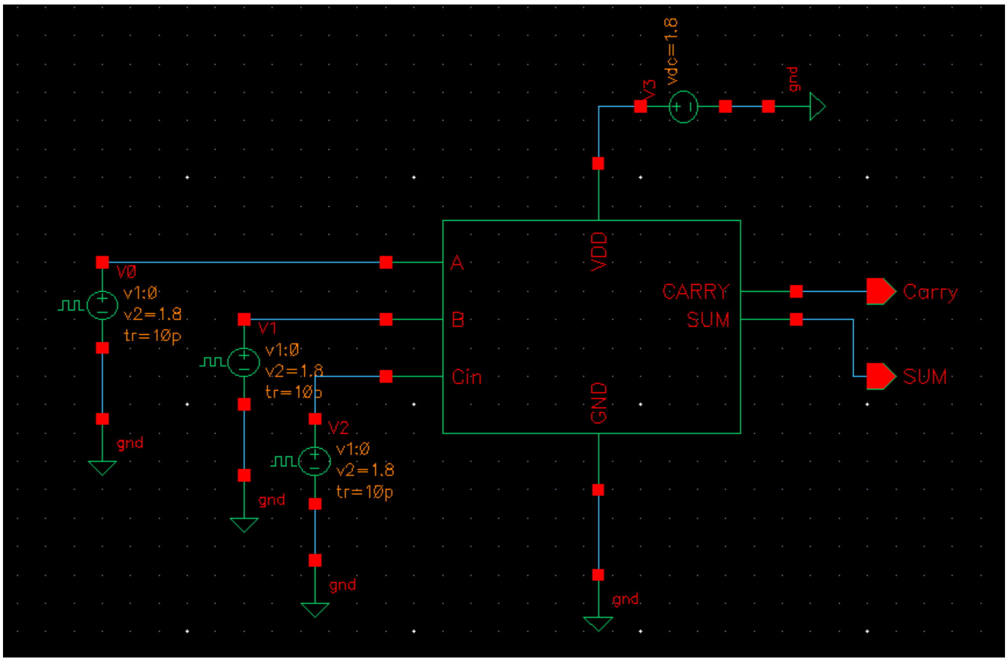

Figure 8: Full Adder testbench (inputs A, B, Cin; outputs SUM and CARRY)

5. 2×2 Vedic Multiplier — Top-Level Schematic

The complete multiplier is assembled from the verified building blocks above. Four AND gate symbols generate the partial products a0b0, a1b0, a0b1, a1b1. Two Full Adder instances (visible in the lower portion of the schematic) accumulate the partial products according to the Urdhva Tiryakbhyam algorithm to produce the 4-bit output S0–S3 (p0–p3) and the final carry-out (cout).

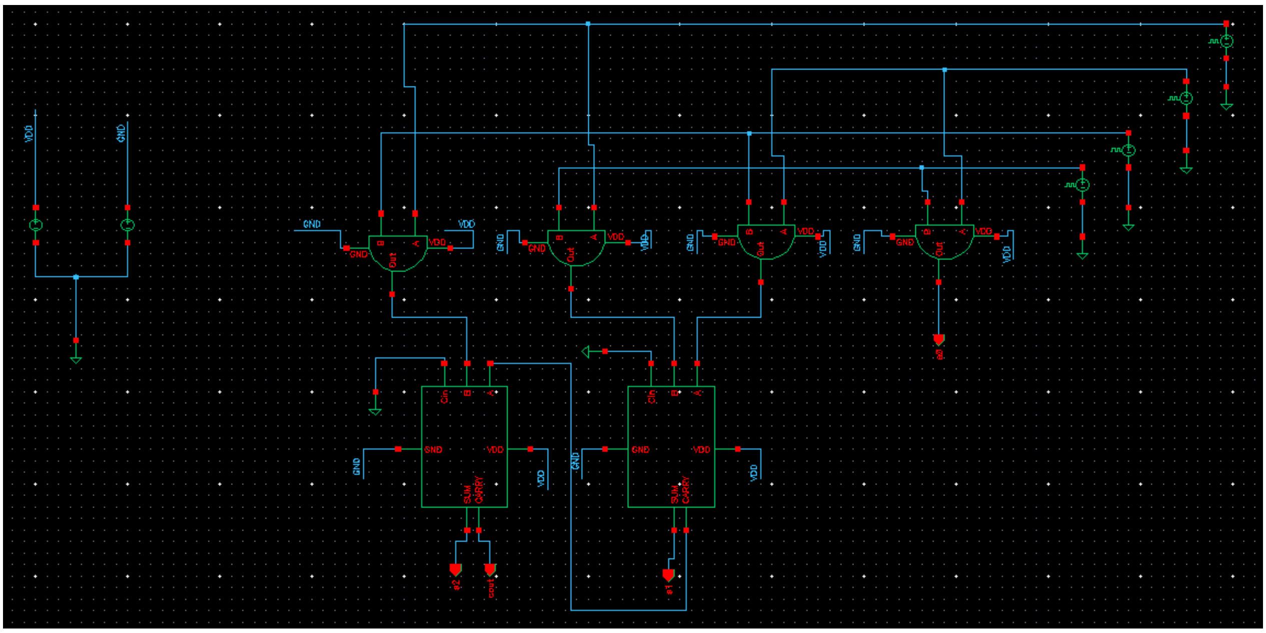

Figure 9: 2×2 Vedic Multiplier — top-level schematic in Cadence Virtuoso (AND gates for partial products + Full Adder tree for accumulation)

Stage	Cells used	Signals produced
Partial products	4× AND gate	a0b0, a1b0, a0b1, a1b1

Bit P[0]	—	a0b0 → S0 (direct)
Bits P[1] + C1	Full Adder #1	S1 = (a1b0 XOR a0b1), C1
Bits P[2] + C2	Full Adder #2	S2 = (a1b1 XOR C1), C2
Bit P[3]	—	S3 = C2 (cout)

Table 3: Multiplier datapath — cell instantiation summary

6. Simulation Results

A transient simulation was run using Cadence Spectre over a 400 ns window. The four input bits (net11, net10, net12, net9 corresponding to a0, a1, b0, b1) are driven by pulse sources with different periods so that all 16 input combinations are exercised within the simulation window. The output waveforms S0 (/s0), S1 (/s1), S2 (/s2), and carry-out (/cout) are captured and verified against the expected multiplication results.

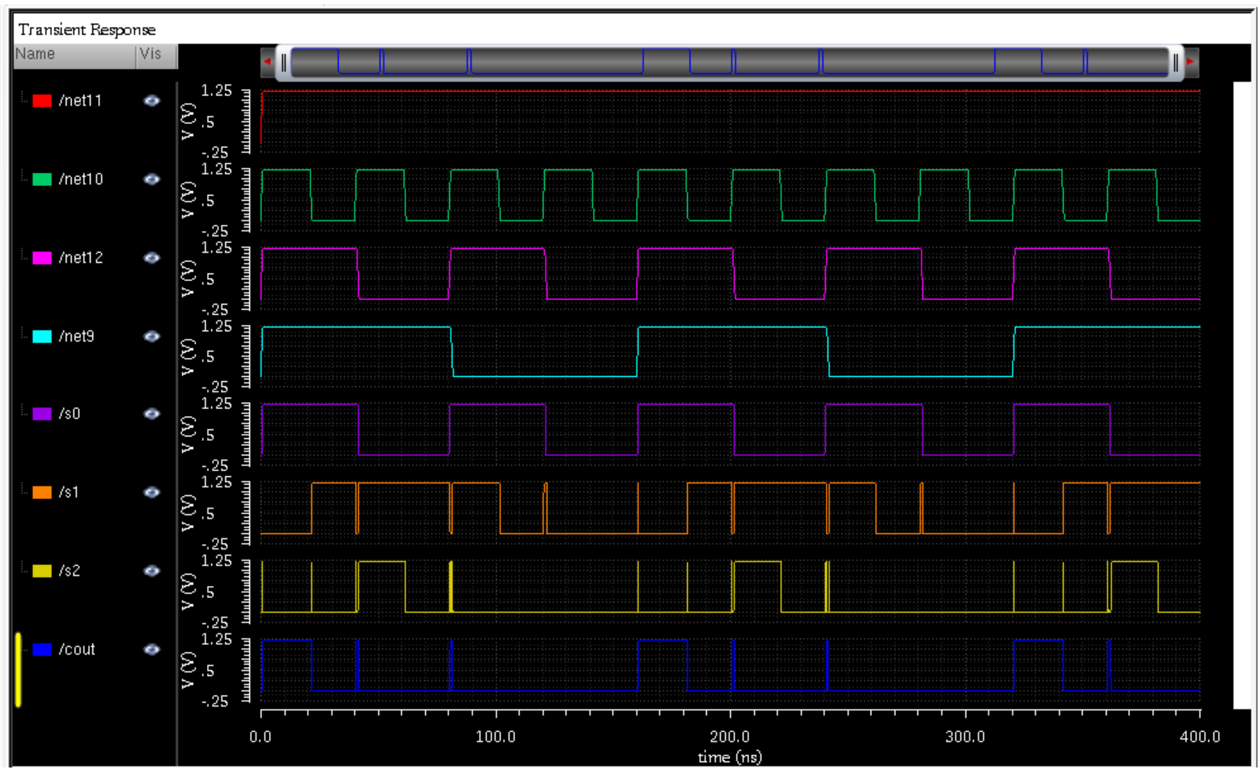

Figure 10: Cadence Spectre transient simulation — input nets (net11, net10, net12, net9) and output product bits S0, S1, S2, cout over 400 ns

The waveforms confirm correct operation for all input combinations. Key observations:

- All output transitions align with the correct multiplication result at each input state.
- S0 (/s0) toggles at the highest rate, consistent with the LSB of the product.
- S1 (/s1) and S2 (/s2) show the expected intermediate toggle patterns.
- Carry-out (/cout) is asserted only for the input combination 11 × 11 = 1001 (decimal 9), where P[3] = 1 — confirming correct carry propagation.
- No glitches or metastability observed across the 400 ns window.

6.1 Verification Summary

Block	Combinations tested	Outputs verified	Result
OR gate	4	Out = A OR B	✓ Pass
AND gate	4	Out = A AND B	✓ Pass
XOR gate	4	Out = A XOR B	✓ Pass
Full Adder	8	SUM, CARRY	✓ Pass
2×2 Multiplier	16	S0–S2, cout	✓ Pass

Table 4: Simulation verification summary

7. Performance Characteristics

Metric	Value / Observation
Supply voltage	1.8 V
Simulation window	400 ns (all 16 combos verified)
Critical path depth	2 logic levels: AND → XOR
Estimated tp	~150–300 ps (90 nm)
Max operating frequency	≥ 1 GHz
Logic gates	4× AND, 2× XOR (6 gates total)
Transistor count	~50–60 (all cells combined)
Power (estimated)	~10–40 µW @ 1 GHz switching

Table 5: Performance characteristics (gpdk090, VDD = 1.8 V)

8. Conclusion

A 2×2 Vedic multiplier was successfully designed from transistors upward in Cadence Virtuoso using the gpdk090 90 nm CMOS PDK. Each sub-cell (OR, AND, XOR, Full Adder) was independently verified before hierarchical integration. The complete multiplier schematic was simulated in Cadence Spectre and confirmed correct operation across all 16 input combinations in a 400 ns transient run. The two-level critical path of the Urdhva Tiryakbhyam architecture makes this design highly suitable for high-speed, low-power arithmetic datapaths. Future work includes physical layout, post-layout parasitic extraction, and scaling to a 4×4 or 8×8 Vedic multiplier.